

Executive function mediates the effects of genetic liability to schizophrenia on behavior and functioning in a community sample of children and adolescents

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ABSTRACT

Background: Genetic liability for schizophrenia has been associated with cognitive deficits and clinical phenotypes during neurodevelopment. However, the possible role of executive function (EF) as a mediator between the polygenic risk score for schizophrenia (PRS-SZ) and subsequent outcomes, including transdiagnostic general psychopathology (the p-factor), has yet to be examined. In this study, we investigated whether EF mediates the effect of PRS-SZ on psychopathology and functional phenotypes in a community-based youth sample.

Study design: We analyzed cross-sectional data from 698 participants (aged 6–14) of the Brazilian High-Risk Cohort. PRS-SZ was calculated using summary statistics from the Psychiatric Genetics Consortium (PGC-3). EF was assessed through working memory, inhibitory control, and time processing tasks, condensed into a single latent EF trait. Independent linear models were tested to examine direct associations between PRS-SZ and EF, general psychopathology, anxiety symptoms, psychotic experiences (PE), and school performance. Mediation models were applied to evaluate EF as a mediator of the associations between PRS-SZ and each outcome.

Study results: PRS-SZ predicted lower EF scores ($\beta = -0.091$, $t = -2.435$, $p = 0.015$). No direct associations were found between PRS-SZ and the other measures. EF mediated the association between PRS-SZ and general psychopathology (p-factor) (Effect = 0.0079, BootSE = 0.0049, LLCI = 0.0004, ULCI = 0.0191), anxiety symptoms (Effect = 0.0075, BootSE = 0.0047, LLCI = 0.0001, ULCI = 0.0182), and school performance (Effect = -0.0167 , BootSE = 0.0077, LLCI = -0.0327 , ULCI = -0.0028), but not PE.

Conclusions: PRS-SZ was associated with poorer EF, which in turn mediated its associations with increased general psychopathology (p-factor) and anxiety symptoms, and reduced school performance in this community youth sample. EF may represent a hub through which PRS-SZ contributes to negative behavioral and functional outcomes.

1. Introduction

For the past three decades, schizophrenia has been recognized as a neurodevelopmental disorder (Murray and Lewis, 1987; Weinberger,

1987). However, the phenotypic trajectories from genetic susceptibility to the full-blown disease across development remain largely unexplored. Examining how genetic liability to schizophrenia and related phenotypes interplay in youths may provide critical insights into this

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neurodevelopmental period, which would be fundamental for improving diagnostic and preventive strategies.

Early mild cognitive and behavioral abnormalities are often observed in individuals who later develop schizophrenia. Whereas premorbid impairments tend to be global in schizophrenia (Meier et al., 2014; Mohn-Haugen et al., 2022), evidence suggests deficits in specific cognitive domains, such as verbal abilities and executive function (EF) (Sheffield et al., 2018), as well as academic performance and social functioning (Cannon et al., 2000; Erlenmeyer-Kimling et al., 2000; Hans et al., 1999; David et al., 1997; Jones et al., 1994). Additionally, early manifestations of anxiety and depressive symptoms during childhood have also been associated with the later emergence of psychosis (Freeman and Garety, 2003; van Os et al., 2014). These findings, nonetheless, are derived mainly from retrospective studies of clinical patients.

Among community samples of children and adolescents, psychotic experiences (PE) and polygenic risk scores for schizophrenia (PRS-SZ) are widely used metrics to investigate psychosis proneness. PE are mild, subthreshold psychotic symptoms, reported by approximately 5–10% of the general population (Linscott and van Os, 2013). They rest on a lower range of the psychosis continuum, thus serving as a possible tool to understand psychotic disorders (Dağdeviren Ertas, 2023; Poulton et al., 2000). In turn, the polygenic risk score summarizes each genetic risk variant for schizophrenia, which individually has a small effect, into a single quantitative index, allowing for the exploration of possible impact of a cumulative genetic risk on related phenotypes (Birnbaum and Weinberger, 2017; Purcell et al., 2009; Consortium SWGotPG, 2014).

The presence of PE has been linked to higher rates of general psychopathology (Rössler et al., 2011), impaired global functioning (Kelleher et al., 2015), and increased suicidal behavior and substance abuse (Kelleher et al., 2014; Cederlöf et al., 2017), even in the absence of a psychotic disorder. In addition, worse cognitive performance has been found in subjects presenting PE (Kelleher et al., 2013; Satterthwaite et al., 2016), even though most studies failed to find direct associations between PRS-SZ and PE during childhood and adolescence (Zammit et al., 2014; Sieradzka et al., 2014; Jones et al., 2016). Meanwhile, significant associations have been reported between PRS-SZ and anxiety symptoms (Jones et al., 2016) and more internalizing problems (Jansen et al., 2018), suggesting that, during earlier developmental stages, behavioral expressions of genetic liability to schizophrenia may manifest as non-psychotic unspecific symptoms.

Studies addressing behavioral correlates to the genetic liability to schizophrenia achieved more promising results with cognitive phenotypes, particularly intelligence quotient (IQ) (Germeine et al., 2016; McIntosh et al., 2013; Riglin et al., 2017). In parallel, the effect of PRS-SZ on school performance seems particularly robust, being reported in at least three independent samples (Jansen et al., 2018; Krapohl et al., 2016; Le Hellard et al., 2017; Koch et al., 2021). Despite the findings on PRS-SZ and cognition, few studies have addressed the association between EF, a central cognitive impairment in schizophrenia, and PRS-SZ among children and adolescents (Riglin et al., 2017; Koch et al., 2021; Manfro et al., 2021; Askeland et al., 2022).

EF encompasses several higher-order cognitive abilities that underlie goal-directed behavior (Miller and Cohen, 2001). The most widely accepted model of EF proposes a division into three functions: shifting, updating (also termed “working memory”), and inhibitory control (Miyake et al., 2000). Interestingly, a single factor model of executive function provided the best model fit in children (Brydges et al., 2012). The same variable was shown to be distinguishable, although correlated, with IQ (van Aken et al., 2016). EF represents a logical target to understand the connection between genetic liability for schizophrenia and phenotypes, considering both its centrality on daily adjustment to environmental demands and its consolidated association with schizophrenia (Sheffield et al., 2018). From this perspective, the interplay between psychopathology, cognition, and functional outcomes can be seen as a network of interacting and reinforcing elements (Borsboom

and Cramer, 2013). However, no previous study investigated the association of a single EF variable with PRS-SZ in children and adolescents and its possible mediator role for further clinical measures.

We hypothesize that, in earlier stages of development, genetic liability to schizophrenia more directly affects cognition, a marker of brain functioning, and indirectly affects other behavioral and functional outcomes. Thus, we aimed to investigate whether the PRS-SZ associates with EF, and whether EF mediates the association between PRS-SZ and behavioral/functional outcomes, namely general psychopathology, anxiety symptoms, PE, and school performance, in a community-based sample of children and adolescents.

2. Methods

2.1. Participants

The sample consisted of children and adolescents aged 6 to 14 from the Brazilian High-Risk Cohort (BHRC), recruited from 57 public schools (22 in Porto Alegre and 35 in São Paulo) (Salum et al., 2015). Briefly, 8012 families (9937 eligible children, 45,394 family members) were interviewed with the Family History Screen (FHS) (Weissman et al., 2000). An index of family load was computed for each of the potential eligible children based on the percentage of members in the family that screened positively for each of the psychiatric disorders assessed, adjusted for relatedness. After screening, 2511 children were fully assessed, 62% considered high-risk for psychiatric disorders and 38% randomly selected. More details about the study design and sampling procedures can be found in Salum et al. (Salum et al., 2015). For this particular study, we utilized the baseline data collected between 2009 and 2010, and only subjects with an IQ of 70 or higher were selected. Written or verbal consent was obtained from the parents or legal guardians of all subjects. The Ethics Committees from the Federal University of São Paulo (CAAE: 74563817.7.2002.5505), University of São Paulo (IORG0004884), and Hospital de Clínicas de Porto Alegre (CAAE: 74563817.7.1001.5327) approved the study.

2.2. Executive function (EF)

Trained psychologists and speech therapists evaluated EF over four sections, including tests on: 1) working memory (the digit span subtest of the Wechsler Intelligence Scale for Children WISC-III (Cruz, 2005) and the Corsi blocks task (Vandierendonck et al., 2004)); 2) inhibitory control (the conflict control (Hogan et al., 2005) and the go/no-go tasks (Bitsakou et al., 2008)); and 3) time processing (the 400-ms and 2000-ms time anticipation tasks (Toplak and Tannock, 2005)). We employed a previously published EF second order model (Fig. S1), which was developed in this sample using confirmatory factor analysis (CFA) (Martel et al., 2017), and yielded excellent fit to the data.

2.3. General psychopathology

In order to assess general psychopathology, we used a latent variable previously generated and validated within our sample (Martel et al., 2017). Parents answered the Development and Well-Being Assessment (DAWBA) (Goodman et al., 2000), a structured interview with a trans-diagnostic approach. The questionnaire targeted current issues that caused significant distress or social impairment. Diagnoses were established using diagnostic probabilistic bands, also reviewed by clinicians, with good concordance. The P-factor structural model of psychopathology was tested through confirmatory factor analyses of the DAWBA data, and three specific factors – fear, distress, and externalizing – exhibited the best fit (Martel et al., 2017).

2.4. Anxiety symptoms

Anxiety symptoms were measured through the Screen for Child

Anxiety Related Emotional Disorders (SCARED) (Birmaher et al., 1997), a 41-item scale that encompasses five dimensions of anxiety symptoms (panic/somatic, generalized anxiety, separation anxiety, social phobia and school phobia). A Portuguese version of the SCARED has been previously validated in Brazilian youths (Isolan et al., 2011). In our study, we generated a single latent variable for the SCARED scores using a CFA of five dimensions, as previously validated in a Brazilian sample (Isolan et al., 2011), which yielded excellent fit to the data (Fig. S2).

2.5. Psychotic experiences (PE)

PE was assessed using the positive dimension of the Community Assessment of Psychotic Experiences (CAPE) (Konings et al., 2006; Mark and Touloupoulou, 2016). The scale consists of 20 questions regarding perceptual abnormalities, thought insertion and broadcasting, persecutory delusions, bizarre experiences, magical thinking and grandiosity. Considering the age range of our sample (6 to 14 years old), we opted for a clinical impression variable: trained psychologists read each item to the subject and evaluated the responses after examining the clinical relevance in light of their cultural background, age, and the degree of conviction related to the experience. Children provided yes/no responses to each item, after which psychologists classified the likelihood that the experience was psychotic as (1) improbable, (2) unlikely, (3) very likely, or (4) certain. We used a single latent variable for the CAPE scores, generated in a previous study from our group using hierarchical CFA, with goodness-of-fit measures (Pan et al., 2019).

2.6. General functioning

2.6.1. School performance

We used the Child Behavior Checklist (CBCL – School) (Ivanova et al., 2007) to assess school performance, through caregiver-report. Possible scores were “failing”, “below average”, “average”, and “above average” on standard educational subjects (Portuguese/Literature, History/Social studies, English/Spanish, Mathematics, Biology, Sciences, Geography and Computer studies). Here we employed saved factor scores from the CFA model performed by Hoffmann et al. (Ivanova et al., 2007; Hoffmann et al., 2016).

2.7. Genotyping and polygenic risk scoring

Genomic DNA from saliva using prepIT-L2P reagent (DNAGENOTEK), we genotyped samples using the Illumina Global Screening Array (GSA), which includes approximately 700,000 single nucleotide polymorphisms (SNPs) markers. Quality control was performed in PLINK v1.9 (Purcell et al., 2007). SNPs were removed if they had a minor allele frequency <1%, a genotype call rate <90%, or violated Hardy-Weinberg equilibrium ($p < 1 \times 10^{-6}$). We also excluded samples with a call rate of <90%, sex discrepancies, heterozygosity outliers (>3 SD from the mean) and relatedness above an identity by descent of 0.12. After applying sample quality control, the final dataset consisted of 1733 participants. Principal components analysis (PCA) was also performed in PLINK v1.9 and the first 10 components were included as covariates in downstream analyses.

To generate the PRS-SZ we used the default options of the PRSice software v.2.3.5 (www.PRSice.info) (Choi and O'Reilly, 2019). Briefly, we performed p -value-informed clumping with a cut-off of $r^2 = 0.1$ within a 250 kb window excluding those SNPs in linkage disequilibrium. The summary statistics from PGC-3 Schizophrenia (downloaded from <https://www.med.unc.edu/pgc>) was set as a training sample (Trubetskoy et al., 2022) and our genomic non-imputed samples as targets. The approach used for PRS calculation follows the method outlined in Purcell et al. (Purcell et al., 2009).

Global ancestry proportions were estimated using ADMIXTURE (K = 3) in unsupervised mode (Alexander et al., 2009). The BHRC dataset was merged with external reference individuals selected to reflect the

primary ancestral components of the Brazilian population: Europeans, Africans, and Native Americans from the combined 1000 Genomes + HGDP callset (Koenig et al., 2024).

Considering that Brazil has a high level of admixture and that the PGC-3 sample consists of mostly Europeans or East Asian participants, we included only individuals with European ancestry above 70% ($N = 698$), although cut-offs of 50%, 60% and 80% have also been tested. Last, the PRSice calculated PRS at distinct p -value thresholds, and after testing various thresholds in our linear models, we adopted $p < 0.05$ for all subsequent analysis, which showed the largest effect of PRS-SZ on the EF variable. All tests can be found in the Supplementary material – Tables S1–S5.

2.8. Statistical analyses

We evaluated our data following a predefined sequence. Initially, we built linear models to explore the direct relation between genetic liability to schizophrenia (PRS-SZ) and 1) executive function; 2) general psychopathology; 3) anxiety symptoms; 4) psychotic experiences; and 5) functioning (school performance). We included age, sex, study site (São Paulo or Porto Alegre) and socioeconomic status as covariates in all analyses, as well as the 10 principal components of the genotyping as covariates. Socioeconomic status was assessed according to a national official instrument (The Economic Classification Criterion Brazil) (ABEP, 2008), which takes purchase capacity and level of education of the head-of-household, and stratified into four groups.

Following the same sequence for exploring the linear associations, we created four independent mediation models with PRS-SZ as the predictor, the latent EF variable as mediator and the other sets of variables as outcomes (Fig. 1).

All analyses were performed using SPSS v.26 and PROCESS Macro for SPSS 4.0. P -value was set at 5%. The indirect effect was estimated using 10,000 bootstrap samples for bias corrected at 95% confidence interval for mediation analyses, as suggested by Preacher and Hayes (Preacher and Hayes, 2004). Confidence intervals that do not include zero indicate significant effects.

3. Results

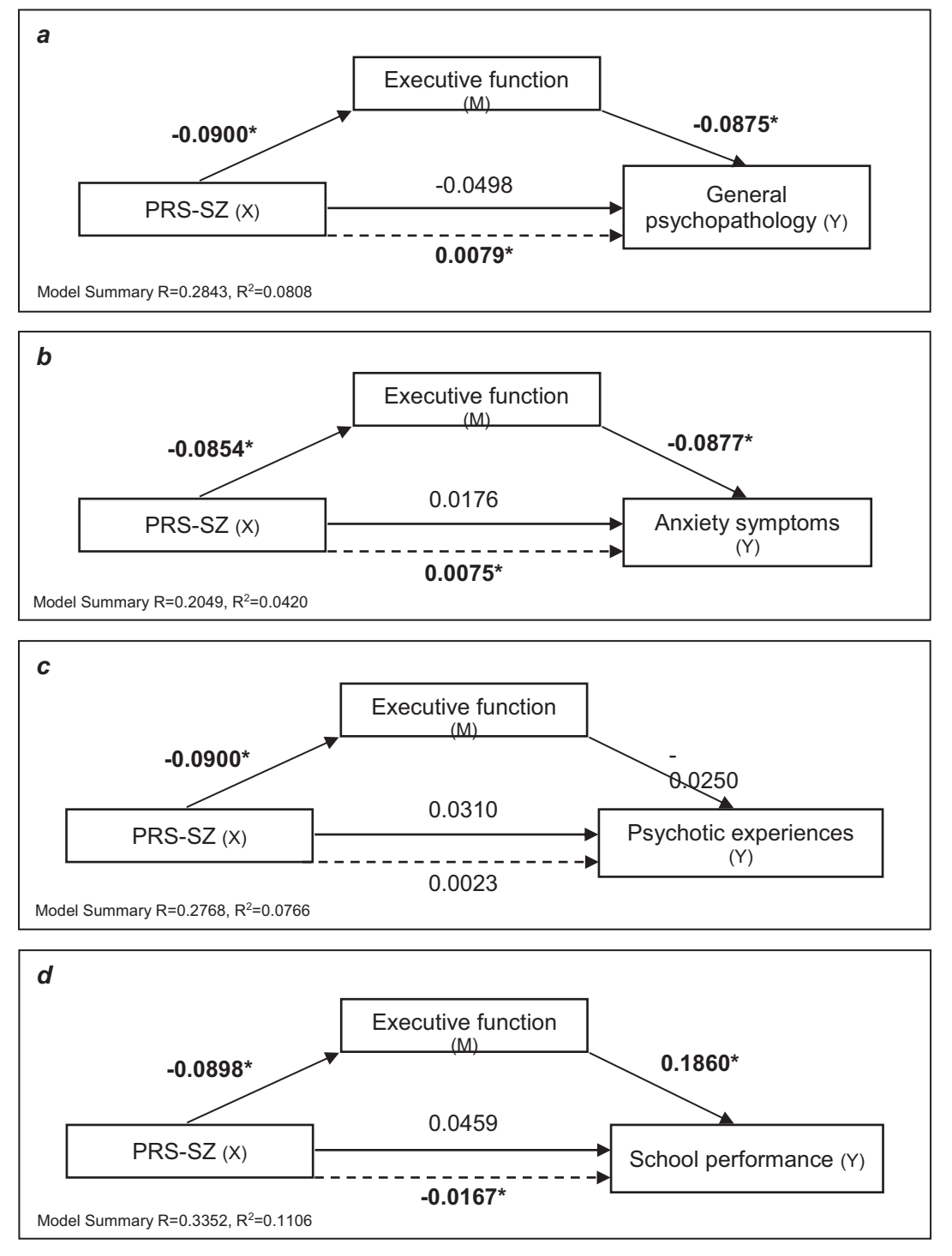
The final sample consisted of 698 individuals, 434 (62.2%) from Porto Alegre and 264 (37.8%) from São Paulo. The mean age was 10.2 years ($SD = 1.96$, Range = 6–14), and 54.3% ($n = 379$) were male. IQs ranged from 70 to 156 (mean = 106, $SD = 16.3$). Socioeconomic status groups were as follows: 24.9% ($n = 174$) of the participants as class A/B (highest strata), 66.2% ($n = 462$) as class C and 8.9% ($n = 62$) as class D/E (lowest strata). The demographic data and following results regard the individuals with European ancestry above 70%. Additional cut-offs can be found in the Supplemental material – Table S5.

3.1. Linear models

We performed independent linear regressions for each outcome, including age, sex, study site, and socioeconomic status as covariates. PRS-SZ significantly predicted lower scores on the latent EF measure ($\beta = -0.091$, $t = -2.435$, $p = 0.015$, $R^2 = 0.047$). None of the other models yielded significant results: general psychopathology ($\beta = -0.043$, $t = -1.161$, $p = 0.246$, $R^2 = 0.060$), anxiety symptoms ($\beta = 0.026$, $t = 0.689$, $p = 0.491$, $R^2 = 0.023$), PE ($\beta = 0.032$, $t = 0.874$, $p = 0.383$, $R^2 = 0.070$), or school performance ($\beta = 0.029$, $t = 0.790$, $p = 0.430$, $R^2 = 0.064$). The results were unchanged with the removal of covariables.

3.2. Mediation models

We created four simple mediation models for each of our initial interest outcomes, all using EF as mediator. Additionally, all models were adjusted for several potential confounders, such as sex, age, and

Mediation diagrams

--> Indicates indirect effects

* Significant effects

Fig. 1. Mediation models testing the indirect effects of polygenic risk score for schizophrenia (PRS-SZ) on four outcomes via executive function (EF). EF mediated the associations between PRS-SZ and (a) general psychopathology, (b) anxiety symptoms, and (d) school performance, as indicated by significant indirect effects. In contrast, no significant mediation or direct effect was observed in the model with (c) psychotic experiences (PE) as the outcome.

socioeconomic status.

We found significant indirect effects linking PRS-SZ to general psychopathology (Effect = 0.0079, Bootstrapped S.E. [Boot SE] = 0.0049, Bootstrapped Lower Confidence Interval (LLCI) = 0.0004, Bootstrapped Upper Confidence Interval (ULCI) = 0.0191; Fig. 1a), and PRS-SZ to anxiety symptoms (Effect = 0.0075, Boot SE = 0.0047, LLCI = 0.0001, ULCI = 0.0182; Fig. 1b), both via EF, as a full mediator.

The model assessing our functional outcome also proved to be a full mediation, yielding significant results for the indirect pathway between PRS-SZ and school performance (Effect = -0.0167, Boot SE = 0.0077, LLCI = -0.0327, ULCI = -0.0028; Fig. 1d). However, in contrast to our initial hypothesis, we found no direct or indirect significant effects in any of the models which included PE as the outcome (Effect = 0.0023, Boot SE = 0.0042, LLCI = -0.0059, ULCI = 0.0113; Fig. 1c), since the LLCI and ULCI interval did include zero.

Mediation diagrams with effects can be seen in Fig. 1. Detailed results can be found in Supplementary Tables S1–S4.

4. Discussion

Our results support the concept that EF represents a hub to understand the impact that genetic vulnerability for schizophrenia might perform on several behavioral and functional phenotypes in a school-based community sample. Secondary findings help contextualize the observed pattern of results, including: 1) the effect of PRS-SZ on EF, but on no other variables in the linear models; 2) the effect of PRS-SZ on general psychopathology and anxiety symptoms, mediated by EF; 3) the effect of PRS-SZ on school performance, a functional outcome, mediated by EF; and 4) the lack of a significant direct or indirect association between PRS-SZ and PE in children and adolescents.

Previous studies showed that PRS-SZ may impact several cognitive measures, mostly through IQ, in community samples (Germin et al., 2016; McIntosh et al., 2013; Riglin et al., 2017; Hubbard et al., 2016). Our results suggest that the link between PRS-SZ and IQ may arise from the effects of underlying cognitive skills, like EF, which may explain the reported associations between premorbid cognitive function and risk for schizophrenia. We decided to use a latent EF measure that encompasses results from different cognitive tasks with a good model fit, reducing noise of individual tasks variables. In addition, we adopted a unitary approach to EF, instead of specific subdomains (like working memory), because these subdomains might not be independent yet in children's developing brain (Brydges et al., 2012). As such, we believe a general common factor represents a more reliable measure, thus improving the interpretability of the results.

We found that PRS-SZ, mediated by EF, has an impact on the severity of anxiety symptoms. Such a result suggests a potential pathway through which genetic liability to schizophrenia may overlap with vulnerability to other psychiatric disorders, possibly mediated by stress susceptibility, a transdiagnostic factor whose impact depends on the timing and intensity of exposure (Grace, 2016; Lee et al., 2013; Gomes et al., 2019). Also, it possibly encourages further research on common pathophysiologicals for related disorders. Seemingly, performing poorly on a cognitive skill essential for adequate social behavior and functional demands, like EF, may be a marker of an underlying biological process more specific to psychosis, as well as a booster to current and future difficulties in different psychiatric disease trajectories. In fact, mPFC disruption can lead to disrupted control of the amygdala (i.e., less control of anxiety), leading to increased susceptibility to stress-induced pathology (Rosenkranz and Grace, 2002; Uliana et al., 2020; Gomes and Grace, 2017a).

A recent genetic study found positive correlations between schizophrenia and all other psychiatric phenotypes analyzed (Duncan et al., 2018). Less specific symptom presentation seems to underlie the genetic liability to schizophrenia at the age range of our sample. These results confirm previous findings in two ways: 1) most published data on PRS-SZ and PE were non-significant (Zammit et al., 2014; Sieradzka et al.,

2014; Jones et al., 2016); 2) birth cohorts have reported that anxiety and other unspecific symptoms are common in the same age range as our study, in individuals who later developed schizophrenia (Gomes and Grace, 2017a; Zimmerman et al., 2015; Gomes and Grace, 2017b; Welham et al., 2009; Johnstone et al., 2000). At this point, cohort studies should help us determine whether EF correlates with psychotic symptoms in older individuals, especially in those who develop psychotic disorders later on.

The lack of significant associations between PE and PRS-SZ may represent, in fact, the absence of a biological link between the PE phenotype and the liability to schizophrenia, or else simply underscore the fragility of the PE construct as currently measured in children and adolescents. To deal with this second issue, we selected a latent variable of PE, obtained from a well-validated instrument (CAPE) using a multilevel structure, which yielded a good model fit (Konings et al., 2006). Recently, we showed that PE (both self-report and clinically validated) and at-risk-mental state overlap in a continuum of severity, with subjects at-risk depicting higher levels of general psychopathology and fewer positive attributes (Moriyama et al., 2018).

The findings on how PRS-SZ may affect school performance in a community sample is relevant in at least three ways: 1) school performance represents the strongest predictor of work and social adjustment in adult life (Heyman et al., 2014; Kirkcaldy et al., 2004); 2) the recognition of the underlying cognitive skill associated may contribute to better screening strategies; 3) the identification of the role of EF may boost preventive programs. Our results aligns with previous findings of PRS-SZ associated with school performance (Jansen et al., 2018; Krapohl et al., 2016; Le Hellard et al., 2017), and highlight EF as one of its mediators.

There are several limitations that should be considered in interpretation of these findings. First, the cross-sectional assessment does not allow determining causation. Despite the use of mediation analyses and the findings that emerged, our current impression is of genetic liability “dilution” across proximal and distal phenotypes on a specific age range rather than a causal model. EF deficits are increasingly recognized as a consequence of heightened stress susceptibility (Gomes and Grace, 2017a; Perica and Luna, 2023). Building upon these findings, future studies could investigate causal inferences by targeting brain networks related to EF, further elucidating the mechanisms through which stress impacts EF disruption. Another limitation concerns the admixture of the sample, characteristic of Brazil, which may impact the interpretation of PRS-SZ (Martin et al., 2017). Our subjects were assessed in two cities with similar ancestry backgrounds (including European, African and Native American contribution) but distinct patterns of admixture. Although both sites have a majority of European contribution (~70%), we observe greater African contribution in São Paulo than in Porto Alegre. However, results were mainly the same when cut-offs of 50% and 60% of European ancestry were used (Table S5). We hypothesized that no effects were found on the 80% cut-off due to lower number of subjects included, which hinders chances of finding significant associations.

Also, our executive function assessment did not include variables on other cognitive skills that compose EF (cognitive flexibility, for example), which should be explored in future studies. Moreover, we cannot rule out that the observed results on EF could be better explained by more basic and specific cognitive skills. Lastly, the effect sizes on the mediation models were very low, though this would be expected considering our non-clinical sample. Our general objective was to generate data to test our hypothesis. Clearly a wide set of additional increments can be suggested, either by including environmental or other biological measures, i.e. neuroimaging data.

Genetics and environment interact at many levels in complex dynamic loops over development. One question for future studies is whether the poor personal and functional achievements may lead to an increased risk to additional environmental adverse factors. For example, subjects with poor EF may have some disadvantage in cognitive

demanding tasks, therefore boosting self-reference thoughts and perceptions in a susceptible individual. Another major concern raised is that ordinary preventive strategies concentrating on symptoms may be too late. Reinforcing support for people with school difficulties should be a more effective approach, even if not by reducing risk, by promoting better functional outcomes. Another important perspective on this association is bridging the gap between genetic liability to schizophrenia, PE, and functioning.

In sum, our study proposes a broader model to understand the complex relations between genetic vulnerability to schizophrenia, PE, general psychopathology, cognitive skills, and functioning, possibly helping to identify individuals at-risk and improve preventive interventions on psychotic disorders among this age span. These results add new possibilities to investigate the underlying neurobiology of brain trajectories towards psychotic disorders. Additional studies will be required to confirm our results and to investigate the implication and effectiveness of preventive strategies.

CRedit authorship contribution statement

Lais Fonseca: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Lucas Toshio Ito:** Writing – review & editing, Data curation. **Carolina Ziebold:** Writing – review & editing, Formal analysis. **Luis A. Rohde:** Writing – review & editing. **Eurípedes C. Miguel:** Writing – review & editing. **Rodrigo A. Bressan:** Writing – review & editing. **Pedro M. Pan:** Writing – review & editing. **Sintia I. Belangero:** Writing – review & editing. **Marcos L. Santoro:** Writing – review & editing, Writing – original draft, Data curation. **Felipe V. Gomes:** Writing – review & editing. **Anthony A. Grace:** Writing – review & editing. **Ary Gadelha:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

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Declaration of competing interest

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Appendix A. Supplementary data

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References

- ABEP, 2008. Critério Padrão de Classificação Econômica Brasil. Assoc Bras Empres Pesqui, pp. 1–3.
- Alexander, D.H., Novembre, J., Lange, K., 2009. Fast model-based estimation of ancestry in unrelated individuals. *Genome Res.* 19 (9), 1655–1664. <https://doi.org/10.1101/gr.094052.109>.
- Askeland, R.B., Hannigan, L.J., Ask, H., et al., 2022. Early manifestations of genetic risk for neurodevelopmental disorders. *J. Child Psychol. Psychiatry* 63 (7), 810–819. <https://doi.org/10.1111/jcpp.13528>.
- Birmaher, B., Khetarpal, S., Brent, D., et al., 1997. The Screen for Child Anxiety Related Emotional Disorders (SCARED): scale construction and psychometric characteristics. *J. Am. Acad. Child Adolesc. Psychiatry* 36 (4), 545–553. <https://doi.org/10.1097/00004583-199704000-00018>.
- Birnbaum, R., Weinberger, D.R., 2017. Genetic insights into the neurodevelopmental origins of schizophrenia. *Nat. Rev. Neurosci.* 18 (12), 727–740. <https://doi.org/10.1038/nrn.2017.125>.
- Bitsakou, P., Psychogiou, L., Thompson, M., et al., 2008. Inhibitory deficits in attention-deficit/hyperactivity disorder are independent of basic processing efficiency and IQ. *J. Neural Transm. (Vienna)* 115 (2), 261–268. <https://doi.org/10.1007/s00702-007-0828-z>.
- Borsboom, D., Cramer, A.O., 2013. Network analysis: an integrative approach to the structure of psychopathology. *Annu. Rev. Clin. Psychol.* 9, 91–121. <https://doi.org/10.1146/annurev-clinpsy-050212-185608>.
- Brydges, C.R., Reid, C.L., Fox, A.M., et al., 2012. A unitary executive function predicts intelligence in children. *Intelligence* 40 (5), 458–469. <https://doi.org/10.1016/j.intell.2012.05.006>.
- Cannon, T.D., Bearden, C.E., Hollister, J.M., et al., 2000. Childhood cognitive functioning in schizophrenia patients and their unaffected siblings: a prospective cohort study. *Schizophr. Bull.* 26 (2), 379–393. <https://doi.org/10.1093/oxfordjournals.schbul.a033460>.
- Cederlöf, M., Kuja-Halkola, R., Larsson, H., et al., 2017. A longitudinal study of adolescent psychotic experiences and later development of substance use disorder and suicidal behavior. *Schizophr. Res.* 181, 13–16. <https://doi.org/10.1016/j.schres.2016.08.029>.
- Choi, S.W., O'Reilly, P.F., 2019. PRSice-2: Polygenic Risk Score software for biobank-scale data. *Gigascience* 8 (7). <https://doi.org/10.1093/gigascience/giz082>.
- Consortium SWGoTPG, 2014. Biological insights from 108 schizophrenia-associated genetic loci. *Nature* 511 (7510), 421–427. <https://doi.org/10.1038/nature13595>.
- Cruz, M., 2005. WISC III: Escala de Inteligência Wechsler para crianças: manual. *Aval. Psicol.* 4, 199–201.
- Dağdeviren Ertaş, B., 2023. Öğretmen Adaylarının Öğretmenlik Mesleğine Yönelik Tutumları, Akademik Güdülenmeleri ve Akademik Özyeterliklerinin CHAID Analizi ile Belirlenmesi. *Nevşehir Hacı Bektaş Veli Üniversitesi SBE Dergisi* 13. <https://doi.org/10.30783/nevsosbilen.1273843>.
- David, A.S., Malmberg, A., Brandt, L., et al., 1997. IQ and risk for schizophrenia: a population-based cohort study. *Psychol. Med.* 27 (6), 1311–1323. <https://doi.org/10.1017/s0033291797005680>.
- Duncan, L.E., Shen, H., Ballon, J.S., et al., 2018. Genetic correlation profile of schizophrenia mirrors epidemiological results and suggests link between polygenic and rare variant (22q11.2) cases of schizophrenia. *Schizophr. Bull.* 44 (6), 1350–1361. <https://doi.org/10.1093/schbul/sbx174>.
- Erlenmeyer-Kimling, L., Rock, D., Roberts, S.A., et al., 2000. Attention, memory, and motor skills as childhood predictors of schizophrenia-related psychoses: the New York High-Risk Project. *Am. J. Psychiatry* 157 (9), 1416–1422. <https://doi.org/10.1176/appi.ajp.157.9.1416>.
- Freeman, D., Garety, P.A., 2003. Connecting neurosis and psychosis: the direct influence of emotion on delusions and hallucinations. *Behav. Res. Ther.* 41 (8), 923–947. [https://doi.org/10.1016/s0005-7967\(02\)00104-3](https://doi.org/10.1016/s0005-7967(02)00104-3).
- Germine, L., Robinson, E.B., Smoller, J.W., et al., 2016. Association between polygenic risk for schizophrenia, neurocognition and social cognition across development. *Transl. Psychiatry* 6 (10), e924. <https://doi.org/10.1038/tp.2016.147>.

- Gomes, F.V., Grace, A.A., 2017a. Prefrontal cortex dysfunction increases susceptibility to schizophrenia-like changes induced by adolescent stress exposure. *Schizophr. Bull.* 43 (3), 592–600. <https://doi.org/10.1093/schbul/sbw156>.
- Gomes, F.V., Grace, A.A., 2017b. Adolescent stress as a driving factor for schizophrenia development—a basic science perspective. *Schizophr. Bull.* 43 (3), 486–489. <https://doi.org/10.1093/schbul/sbx033>.
- Gomes, F.V., Zhu, X., Grace, A.A., 2019. Stress during critical periods of development and risk for schizophrenia. *Schizophr. Res.* 213, 107–113. <https://doi.org/10.1016/j.schres.2019.01.030>.
- Goodman, R., Ford, T., Richards, H., et al., 2000. The Development and Well-Being Assessment: description and initial validation of an integrated assessment of child and adolescent psychopathology. *J. Child Psychol. Psychiatry* 41 (5), 645–655.
- Grace, A.A., 2016. Dysregulation of the dopamine system in the pathophysiology of schizophrenia and depression. *Nat. Rev. Neurosci.* 17 (8), 524–532. <https://doi.org/10.1038/nrn.2016.57>.
- Hans, S.L., Marcus, J., Nuechterlein, K.H., et al., 1999. Neurobehavioral deficits at adolescence in children at risk for schizophrenia: the Jerusalem Infant Development Study. *Arch. Gen. Psychiatry* 56 (8), 741–748. <https://doi.org/10.1001/archpsyc.56.8.741>.
- Heyman, G.M., Dunn, B.J., Mignone, J., 2014. Disentangling the correlates of drug use in a clinic and community sample: a regression analysis of the associations between drug use, years-of-school, impulsivity, IQ, working memory, and psychiatric symptoms. *Front. Psychiatry* 5 (70). <https://doi.org/10.3389/fpsy.2014.00070>.
- Hoffmann, M.S., Leibenluft, E., Stringaris, A., et al., 2016. Positive attributes buffer the negative associations between low intelligence and high psychopathology with educational outcomes. *J. Am. Acad. Child Adolesc. Psychiatry* 55 (1), 47–53. <https://doi.org/10.1016/j.jaac.2015.10.013>.
- Hogan, A.M., Vargha-Khadem, F., Kirkham, F.J., et al., 2005. Maturation of action monitoring from adolescence to adulthood: an ERP study. *Dev. Sci.* 8 (6), 525–534. <https://doi.org/10.1111/j.1467-7687.2005.00444.x>.
- Hubbard, L., Tansey, K.E., Rai, D., et al., 2016. Evidence of common genetic overlap between schizophrenia and cognition. *Schizophr. Bull.* 42 (3), 832–842. <https://doi.org/10.1093/schbul/sbv168>.
- Isolan, L., Salum, G.A., Osowski, A.T., et al., 2011. Psychometric properties of the Screen for Child Anxiety Related Emotional Disorders (SCARED) in Brazilian children and adolescents. *J. Anxiety Disord.* 25 (5), 741–748. <https://doi.org/10.1016/j.janxdis.2011.03.015>.
- Ivanova, M.Y., Dobrea, A., Dopfner, M., et al., 2007. Testing the 8-symptom structure of the child behavior checklist in 30 societies. *J. Clin. Child Adolesc. Psychol.* 36 (3), 405–417. <https://doi.org/10.1080/15374410701444363>.
- Jansen, P.R., Polderman, T.J.C., Bolhuis, K., et al., 2018. Polygenic scores for schizophrenia and educational attainment are associated with behavioural problems in early childhood in the general population. *J. Child Psychol. Psychiatry* 59 (1), 39–47. <https://doi.org/10.1111/jcpp.12759>.
- Johnstone, E.C., Abukmeil, S.S., Byrne, M., et al., 2000. Edinburgh high risk study—findings after four years: demographic, attainment and psychopathological issues. *Schizophr. Res.* 46 (1), 1–15. [https://doi.org/10.1016/S0920-9964\(99\)00225-X](https://doi.org/10.1016/S0920-9964(99)00225-X).
- Jones, P., Rodgers, B., Murray, R., et al., 1994. Child development risk factors for adult schizophrenia in the British 1946 birth cohort. *Lancet* 344 (8934), 1398–1402. [https://doi.org/10.1016/S0140-6736\(94\)90569-X](https://doi.org/10.1016/S0140-6736(94)90569-X).
- Jones, H.J., Stergiakouli, E., Tansey, K.E., et al., 2016. Phenotypic manifestation of genetic risk for schizophrenia during adolescence in the general population. *JAMA Psychiatry* 73 (3), 221–228. <https://doi.org/10.1001/jamapsychiatry.2015.3058>.
- Kelleher, I., Clarke, M.C., Rawdon, C., et al., 2013. Neurocognition in the extended psychosis phenotype: performance of a community sample of adolescents with psychotic symptoms on the MATRICS neurocognitive battery. *Schizophr. Bull.* 39 (5), 1018–1026. <https://doi.org/10.1093/schbul/sbs086>.
- Kelleher, I., Cederlöf, M., Lichtenstein, P., 2014. Psychotic experiences as a predictor of the natural course of suicidal ideation: a Swedish cohort study. *World Psychiatry* 13 (2), 184–188. <https://doi.org/10.1002/wps.20131>.
- Kelleher, I., Wigman, J.T., Harley, M., et al., 2015. Psychotic experiences in the population: association with functioning and mental distress. *Schizophr. Res.* 165 (1), 9–14. <https://doi.org/10.1016/j.schres.2015.03.020>.
- Kirkcaldy, B., Furnham, A., Siefen, G., 2004. *The Relationship Between Health Efficacy, Educational Attainment, and Well-Being Among 30 Nations*. Hogrefe & Huber Publishers.
- Koch, E., Nyberg, L., Lundquist, A., et al., 2021. Sex-specific effects of polygenic risk for schizophrenia on lifespan cognitive functioning in healthy individuals. *Transl. Psychiatry* 11 (1), 520. <https://doi.org/10.1038/s41398-021-01649-4>.
- Koenig, Z., Yohannes, M.T., Nkambule, L.L., et al., 2024. A harmonized public resource of deeply sequenced diverse human genomes. *Genome Res.* 34 (5), 796–809. <https://doi.org/10.1101/gr.278378.123>.
- Konings, M., Bak, M., Hanssen, M., et al., 2006. Validity and reliability of the CAPE: a self-report instrument for the measurement of psychotic experiences in the general population. *Acta Psychiatr. Scand.* 114 (1), 55–61. <https://doi.org/10.1111/j.1600-0447.2005.00741.x>.
- Krapohl, E., Euesden, J., Zabaneh, D., et al., 2016. Phenome-wide analysis of genome-wide polygenic scores. *Mol. Psychiatry* 21 (9), 1188–1193. <https://doi.org/10.1038/mp.2015.126>.
- Le Hellard, S., Wang, Y., Witoele, A., et al., 2017. Identification of gene loci that overlap between schizophrenia and educational attainment. *Schizophr. Bull.* 43 (3), 654–664. <https://doi.org/10.1093/schbul/sbw085>.
- Lee, S.H., Ripke, S., Neale, B.M., et al., 2013. Genetic relationship between five psychiatric disorders estimated from genome-wide SNPs. *Nat. Genet.* 45 (9), 984–994. <https://doi.org/10.1038/ng.2711>.
- Linscott, R.J., van Os, J., 2013. An updated and conservative systematic review and meta-analysis of epidemiological evidence on psychotic experiences in children and adults: on the pathway from proneness to persistence to dimensional expression across mental disorders. *Psychol. Med.* 43 (6), 1133–1149. <https://doi.org/10.1017/S0033291712001626>.
- Manfro, A.G., Pine, D.S., Polanczyk, G.V., et al., 2021. Testing the stability and validity of an executive dysfunction classification using task-based assessment in children and adolescents. *J. Am. Acad. Child Adolesc. Psychiatry* 60 (12), 1501–1512. <https://doi.org/10.1016/j.jaac.2020.11.016>.
- Mark, W., Touloupoulou, T., 2016. Psychometric properties of “Community Assessment of Psychic Experiences”: review and meta-analyses. *Schizophr. Bull.* 42 (1), 34–44. <https://doi.org/10.1093/schbul/sbv088>.
- Martel, M.M., Pan, P.M., Hoffmann, M.S., et al., 2017. A general psychopathology factor (P factor) in children: structural model analysis and external validation through familial risk and child global executive function. *J. Abnorm. Psychol.* 126 (1), 137–148. <https://doi.org/10.1037/abn0000205>.
- Martin, A.R., Gignoux, C.R., Walters, R.K., et al., 2017. Human demographic history impacts genetic risk prediction across diverse populations. *Am. J. Hum. Genet.* 100 (4), 635–649. <https://doi.org/10.1016/j.ajhg.2017.03.004>.
- McIntosh, A.M., Gow, A., Luciano, M., et al., 2013. Polygenic risk for schizophrenia is associated with cognitive change between childhood and old age. *Biol. Psychiatry* 73 (10), 938–943. <https://doi.org/10.1016/j.biopsych.2013.01.011>.
- Meier, M.H., Caspi, A., Reichenberg, A., et al., 2014. Neuropsychological decline in schizophrenia from the premorbid to the postonset period: evidence from a population-representative longitudinal study. *Am. J. Psychiatry* 171 (1), 91–101. <https://doi.org/10.1176/appi.ajp.2013.12111438>.
- Miller, E.K., Cohen, J.D., 2001. An integrative theory of prefrontal cortex function. *Annu. Rev. Neurosci.* 24, 167–202. <https://doi.org/10.1146/annurev.neuro.24.1.167>.
- Miyake, A., Friedman, N.P., Emerson, M.J., et al., 2000. The unity and diversity of executive functions and their contributions to complex “Frontal Lobe” tasks: a latent variable analysis. *Cogn. Psychol.* 41 (1), 49–100. <https://doi.org/10.1006/cogp.1999.0734>.
- Mohn-Haugen, C.R., Mohn, C., Larøi, F., et al., 2022. A systematic review of premorbid cognitive functioning and its timing of onset in schizophrenia spectrum disorders. *Schizophr. Res. Cogn.* 28, 100246. <https://doi.org/10.1016/j.scog.2022.100246>.
- Moriyama, T.S., Drukker, M., Gadelha, A., et al., 2018. The association between psychotic experiences and traumatic life events: the role of the intention to harm. *Psychol. Med.* 48 (13), 2235–2246. <https://doi.org/10.1017/S0033291717003762>.
- Murray, R.M., Lewis, S.W., 1987. Is schizophrenia a neurodevelopmental disorder? *Br. Med. J. (Clin. Res. Ed)* 295 (6600), 681–682. <https://doi.org/10.1136/bmj.295.6600.681>.
- Pan, P.M., Gadelha, A., Argolo, F.C., et al., 2019. Childhood trauma and adolescent psychotic experiences in a community-based cohort: the potential role of positive attributes as a protective factor. *Schizophr. Res.* 205, 23–29. <https://doi.org/10.1016/j.schres.2018.06.044>.
- Perica, M.I., Luna, B., 2023. Impact of stress on excitatory and inhibitory markers of adolescent cognitive critical period plasticity. *Neurosci. Biobehav. Rev.* 153, 105378. <https://doi.org/10.1016/j.neubiorev.2023.105378>.
- Poulton, R., Caspi, A., Moffitt, T.E., et al., 2000. Children’s self-reported psychotic symptoms and adult schizophreniform disorder: a 15-year longitudinal study. *Arch. Gen. Psychiatry* 57 (11), 1053–1058. <https://doi.org/10.1001/archpsyc.57.11.1053>.
- Preacher, K.J., Hayes, A.F., 2004. SPSS and SAS procedures for estimating indirect effects in simple mediation models. *Behav. Res. Methods Instrum. Comput.* 36 (4), 717–731. <https://doi.org/10.3758/bf03206553>.
- Purcell, S., Neale, B., Todd-Brown, K., et al., 2007. PLINK: a tool set for whole-genome association and population-based linkage analyses. *Am. J. Hum. Genet.* 81 (3), 559–575. <https://doi.org/10.1086/519795>.
- Purcell, S.M., Wray, N.R., Stone, J.L., et al., 2009. Common polygenic variation contributes to risk of schizophrenia and bipolar disorder. *Nature* 460 (7256), 748–752. <https://doi.org/10.1038/nature08185>.
- Riglin, L., Collishaw, S., Richards, A., et al., 2017. Schizophrenia risk alleles and neurodevelopmental outcomes in childhood: a population-based cohort study. *Lancet Psychiatry* 4 (1), 57–62. [https://doi.org/10.1016/S2215-0366\(16\)30406-0](https://doi.org/10.1016/S2215-0366(16)30406-0).
- Rosenkranz, J.A., Grace, A.A., 2002. Cellular mechanisms of infralimbic and prelimbic prefrontal cortical inhibition and dopaminergic modulation of basolateral amygdala neurons in vivo. *J. Neurosci.* 22 (1), 324–337. <https://doi.org/10.1523/JNEUROSCI.22-01-00324.2002>.
- Rössler, W., Hengartner, M.P., Ajdacic-Gross, V., et al., 2011. Sub-clinical psychosis symptoms in young adults are risk factors for subsequent common mental disorders. *Schizophr. Res.* 131 (1–3), 18–23. <https://doi.org/10.1016/j.schres.2011.06.019>.
- Salum, G.A., Gadelha, A., Pan, P.M., et al., 2015. High risk cohort study for psychiatric disorders in childhood: rationale, design, methods and preliminary results. *Int. J. Methods Psychiatr. Res.* 24 (1), 58–73. <https://doi.org/10.1002/mpr.1459>.
- Satterthwaite, T.D., Wolf, D.H., Calkins, M.E., et al., 2016. Structural brain abnormalities in youth with psychosis spectrum symptoms. *JAMA Psychiatry* 73 (5), 515–524. <https://doi.org/10.1001/jamapsychiatry.2015.3463>.
- Sheffield, J.M., Karcher, N.R., Barch, D.M., 2018. Cognitive deficits in psychotic disorders: a lifespan perspective. *Neuropsychol. Rev.* 28 (4), 509–533. <https://doi.org/10.1007/s11065-018-9388-2>.
- Sieradzka, D., Power, R.A., Freeman, D., et al., 2014. Are genetic risk factors for psychosis also associated with dimension-specific psychotic experiences in adolescence? *PLoS One* 9 (4), e94398. <https://doi.org/10.1371/journal.pone.0094398>.
- Toplak, M.E., Tannock, R., 2005. Tapping and anticipation performance in attention deficit hyperactivity disorder. *Percept. Mot. Skills* 100 (3 Pt 1), 659–675. <https://doi.org/10.2466/pms.100.3.659-675>.

- Trubetskoy, V., Pardiñas, A.F., Qi, T., et al., 2022. Mapping genomic loci implicates genes and synaptic biology in schizophrenia. *Nature* 604 (7906), 502–508. <https://doi.org/10.1038/s41586-022-04434-5>.
- Uliana, D.L., Gomes, F.V., Grace, A.A., 2020. Prelimbic medial prefrontal cortex disruption during adolescence increases susceptibility to helpless behavior in adult rats. *Eur. Neuropsychopharmacol.* 35, 111–125. <https://doi.org/10.1016/j.euroneuro.2020.04.004>.
- van Aken, L., Kessels, R.P., Wingermühle, E., et al., 2016. Fluid intelligence and executive functioning more alike than different? *Acta Neuropsychiatr.* 28 (1), 31–37. <https://doi.org/10.1017/neu.2015.46>.
- van Os, J., Lataster, T., Delespaul, P., et al., 2014. Evidence that a psychopathology interactome has diagnostic value, predicting clinical needs: an experience sampling study. *PloS One* 9 (1), e86652. <https://doi.org/10.1371/journal.pone.0086652>.
- Vandierendonck, A., Kemps, E., Fastame, M.C., et al., 2004. Working memory components of the Corsi blocks task. *Br. J. Psychol.* 95 (Pt 1), 57–79. <https://doi.org/10.1348/000712604322779460>.
- Weinberger, D.R., 1987. Implications of normal brain development for the pathogenesis of schizophrenia. *Arch. Gen. Psychiatry* 44 (7), 660–669. <https://doi.org/10.1001/archpsyc.1987.01800190080012>.
- Weissman, M.M., Wickramaratne, P., Adams, P., et al., 2000. Brief screening for family psychiatric history: the family history screen. *Arch. Gen. Psychiatry* 57 (7), 675–682. <https://doi.org/10.1001/archpsyc.57.7.675>.
- Welham, J., Isohanni, M., Jones, P., et al., 2009. The antecedents of schizophrenia: a review of birth cohort studies. *Schizophr. Bull.* 35 (3), 603–623. <https://doi.org/10.1093/schbul/sbn084>.
- Zammit, S., Hamshere, M., Dwyer, S., et al., 2014. A population-based study of genetic variation and psychotic experiences in adolescents. *Schizophr. Bull.* 40 (6), 1254–1262. <https://doi.org/10.1093/schbul/sbt146>.
- Zimmerman, E.C., Bellaire, M., Ewing, S.G., et al., 2015. Abnormal stress responsivity in a rodent developmental disruption model of schizophrenia. *Neuropsychopharmacology* 40 (3), 791. <https://doi.org/10.1038/npp.2014.288>.